

Vikramaditya Tatke

Lead Analytics Engineer

vikramadityatatke@gmail.com | +44 7377212035 | <https://github.com/vikramaditya-tatke> | London, United Kingdom

Data and Analytics Engineer and team leader with 5 years of experience driving impactful data solutions and database administration. Proficient in collaborating with stakeholders, mentoring team-members and optimizing analytical workloads. Expertise includes data pipelines, cloud platforms and database systems. Recognised for a proactive approach towards problem-solving.

SKILLS

- **Programming & Scripting Languages:** Python (FastAPI, Pydantic, Boto3, REST APIs), Rust, Java,, shell.
 - **Data Engineering Frameworks:** Apache Kafka, Apache Airflow, Dagster, Pandas/Polars, PySpark.
 - **Databases:** ClickHouse, MongoDB, Postgres, DuckDB, Redis.
- **Cloud:** AWS EC2, AWS Lambda, AWS Redshift, AWS Glue, AWS S3, MS Azure Sentinel, Log Analytics, Blob Storage, Event Hubs.
 - **DevOps and Version Control:** Docker, Jira, Git (Bitbucket and GitHub), GitHub Actions, Linux.
 - **Business Intelligence and Data Visualization:** Tableau, Qlik Sense, Evidence, Streamlit, Matplotlib.

CERTIFICATIONS

- AWS Certified Data Engineer - Associate (Issued 2026)**

ClickHouse Certified Developer (Issued 2025)
- MongoDB Associate Developer (Issued 2024)**

Tableau Desktop Specialist (Issued 2020)

WORK EXPERIENCE

- SecurityHQ**

Lead Data Engineer

London, United Kingdom

December 2024–Present

 - Spearheaded the migration of the primary analytical database from a self-hosted, sharded MongoDB cluster to ClickHouse Cloud. This initiative improved analytical query performance by over 1000x and reduced annual platform costs by 75%.
 - Designed and implemented a reusable data ingestion framework and utility library for Apache Airflow, enabling the team to rapidly develop data pipelines for Cyber Security in 2 to 4 weeks.
 - Identified and resolved complex performance issues in analytical workloads by analyzing query execution plans, pinpointing system bottlenecks, and implementing advanced indexing strategies and materialized views.
 - Mentored and guided a team of 5 data engineers while acting as a code reviewer and providing technical expertise and project oversight for developing Airflow DAGs and creating database schemas.
 - Led the integration of agentic AI-driven development into team workflows by developing specialized sub-agents for identification of data structures in API responses, database schema design, and enforcement of data pipeline best practices.
 - Created CI/CD workflows for deploying Airflow DAGs, updating Docker images and Devcontainers using GitHub Actions.
 - Designed an AI driven analytics pipeline using AWS Bedrock that enriched technical records with impact scores and reduced manual triage time by 40%.
 - Engineered a semantic data reconciliation engine using FAISS and Polars to eliminate manual categorization overhead, enabling the automated mapping of unstructured inputs to strict schemas for reliable downstream analytics.
 - Demonstrated stakeholder empathy by collaborating with product managers and security analysts to translate complex technical initiatives into actionable strategic outcomes.
- Data Engineer**

June 2022–December 2024

 - Acted as the SME for MongoDB and set up a sharded cluster in a secure self-hosted environment from scratch in 3 months.
 - Designed and maintained resilient, scalable and high-throughput, data pipeline in Python by leveraging asynchronous programming to reliably stream over 10TB of security event logs weekly.
 - Served as the subject matter expert for large scale database clusters by designing a secure self hosted environment that supported critical business intelligence operations.
 - Automated data integrity monitoring by developing a Tableau dashboard that reconciled source REST API metadata with target table counts, significantly improving data reliability and stakeholder trust.

Advanced Manufacturing Control Systems (AMCS) Group

Limerick, Ireland

Associate Software Support Engineer

February 2022–June 2022

- Optimized SQL Server databases for customers ensuring data integrity and query performance using best practices.
- Monitored CPU utilization, memory usage, and disk I/O in customer environments to identify potential issues proactively.
- Diagnosed and resolved complex database issues, including system errors, performance bottlenecks, and compatibility problems.

Amazon Web Services

Dublin, Ireland

Software Development Engineer 1 Intern

May 2021–November 2021

- Developed a critical component for the DynamoDB meta-data team that promised 99.999999999% availability.
- Migrated client discovery service from Java to Rust to overcome the drawbacks of the garbage collector and target consistent sub-1ms client-side latency.
- Optimized the service by implementing multi-threading, an asynchronous network runtime and data compression over the wire.

PROJECTS

Real-time Event Processing Pipeline with Flink and Kafka

- Designed and implemented a distributed data pipeline for real-time Windows event log processing using Java, Apache Flink, and Kafka. Demonstrated fault-tolerant stream processing with TCP ingestion, JSON parsing, and Kafka integration.

Polars GPU Acceleration Benchmark

- Implemented and benchmarked a high-performance data processing pipeline utilizing Polars with GPU acceleration. Developed comprehensive Jupyter Notebooks to analyse and visualize performance metrics on large datasets.

Streamlit Admin Panel

- Designed and implemented a fully functional admin panel using Streamlit for real-time data visualization and management. Developed Python scripts to handle backend data processing and integration with various data sources. Ensured seamless user interaction and efficient data handling within the application.

QRadar REST API Kafka Data Pipeline

- Developed a data pipeline to integrate QRadar REST API with Kafka for real-time data ingestion and processing. Utilized Python to implement API calls, data transformation, and Kafka producer/consumer logic. Implemented well-structured logging using Python's logging module, ensuring detailed logs are stored efficiently for monitoring and troubleshooting. Logs are stored in a centralized logging system for easy access and analysis.

EDUCATION

National College of Ireland

Dublin, Ireland

Post Graduate Diploma in Science in Data Analytics

September 2018–March 2020

MITSOM College, Pune University

Pune, India

Bachelors in computer applications (BCA)

June 2014–May 2017